

Application Test Report No. 2

Date: 2026.06.14

A. Description of Activities Performed

Environments and devices used for testing:

- **System:** Windows 11, **Browser:** Google Chrome v149.0.7827.104 (Resolution: 1920x1080)
- **System:** Windows 11, **Browser:** Mozilla Firefox v151.0.3 (Resolution: 1920x1080)
- **Build number:** v2.0.2 **dated** 14.06.2026

Application areas and types of tests performed:

Application Area	Types of Tests Performed	Time Spent on Testing
User Interface	Functional tests, unit tests, regression tests	0,5 h
Session Trends Module	Functional tests, unit tests, regression tests	0,5 h
Statistical Analyses	Functional tests, unit tests, regression tests	0,5 h
Monthly/Quarterly Change distribution Module	Functional tests (switching monthly/quarterly modes, selecting month/quarter), unit tests	1 h
Fetching data from NBP API	Functional test, regression tests	0,5 h

B. Summary of All Defects

1. Fixed and retested:

- a) Missing recent dates for analysis (the last available date was 02.06)
- b) Missing request chunking for API queries.
- c) Missing an appropriate error message in case of no internet connection.

2. To do:

- a) Incorrect business logic for currency pairs (Histogram displays two separate series instead of one cross-rate)
- b) Missing validation for same currency selection

c) Interactive controls are not disabled during API data fetching

3. **Risky to fix at the moment:** (none)

C. Detailed Description of Defects

Test 1a

- **Discovered by:** Aliaksandr Yurusau
- **Report in which discovered:** 1
- **Priority:** Highest
- **Repeatability:** Always
- **Browser:** Not applicable
- **Steps to reproduce:**
 1. Open the application in a browser.
 2. Attempt to change the start date to 03.05.2026 for 1 month period.
- **Test result:** It correctly returns the full month's date

Test 1b

- **Discovered by:** Aliaksandr Yurusau
- **Report in which discovered:** 1
- **Priority:** Medium
- **Repeatability:** Always
- **Browser:** Not applicable
- **Steps to reproduce (e.g., Google Chrome):**
 1. Open the application in a browser.
 2. Select any start date that is at least one year earlier than today's date.
 3. Select an analysis period of 1 year.
 4. Open the developer console by pressing F12.
 5. Go to the Network -> Fetch/XHR tab.
- **Test result:** Four requests to the API covering chunks of 93 days or less (e.g., 2025-01-01/2025-04-03 + 2025-04-04/2025-07-05 + 2025-07-06/2025-10-06 + 2025-10-07/2026-01-01). This does comply with **system constraint #13**: *"the system shall be constrained to fetching a maximum of 93 days of historical data per request. If a user selects a period longer than 93 days, the system shall perform request chunking."*

Test 1c

- **Discovered by:** Aliaksandr Yurusau
- **Report in which discovered:** 1
- **Priority:** Low
- **Repeatability:** Always
- **Browser:** Not applicable

- **Steps to reproduce (e.g., Google Chrome):**
 1. Open the application in a browser.
 2. Open the developer console by pressing F12.
 3. Go to the Network tab.
 4. In the throttling dropdown, select "offline" instead of "no throttling".
 5. Change the analysis data (e.g., change the period to 1 week).
- **Test result:** The message No internet connection. Could not fetch data from NBP API. is displayed.

Test 2a

- **Discovered by:** Aliaksandr Yurusau
- **Report in which discovered:** 2
- **Priority:** Highest
- **Repeatability:** Always
- **Browser:** Not applicable
- **Steps to reproduce:**
 1. Open the application in a browser.
 2. Select Monthly and Quarterly Change Distribution
- **Test result:** The histogram displays two separate bars for each percentage range (one representing USD changes and one representing EUR changes). This means the system calculates the distribution of changes for each currency independently against PLN. According to the specification ("calculate cross-rates to display user-selected currency pairs"), the system must calculate a single cross-rate (EUR/USD) first and display a single distribution histogram representing the changes of that combined pair.

Test 2b

- **Discovered by:** Aliaksandr Yurusau
- **Report in which discovered:** 2
- **Priority:** High
- **Repeatability:** Always
- **Browser:** Not applicable
- **Steps to reproduce:**
 1. Open the application in a browser.
 2. Select Monthly and Quarterly Change Distribution
 3. Select the same currency
- **Test result:** The system allows the user to select the same currency (e.g., USD/USD) in both dropdowns. This directly violates the specification constraint: *"The GUI shall prevent the user from selecting the same currency for both inputs in change distribution analysis."*

Test 2c

- **Discovered by:** Aliaksandr Yurusau
- **Report in which discovered:** 2
- **Priority:** Low
- **Repeatability:** Always with slow connection
- **Browser:** Not applicable
- **Steps to reproduce (e.g. Google Chrome):**
 1. Open the application in a browser.
 2. Open developer console
 3. Select Network
 4. In throttling section select 3G instead of no throttling
 5. Change something on the page (e.g. analysis period)
- **Test result:** Most of the controls are blocked excluding navigation at the header are blocked. This violates the GUI constraint: "Simultaneously, all interactive controls in this view shall be temporarily disabled to prevent overlapping requests." The controls should be "grayed out" or blocked until the API fetch is fully completed.

D. Summary of Exit Criteria for All Tests

1. Did we manage to complete the planned tests? Yes, all planned testing activities were successfully completed. We successfully re-tested the defects reported in Test Report No. 1 and performed functional and acceptance testing of the newly integrated Monthly/Quarterly Change Distribution Module. Tests were executed on the target environment (Windows 11, Chrome/Firefox at 1920x1080).

2. Were there any difficulties? No significant technical difficulties were encountered during testing. The external NBP API was stable, which allowed for the effective use of browser throttling (3G simulation) to uncover the defect related to interactive controls not being properly blocked.

3. Verification of work time estimates. The time spent on testing (totaling 3,5 hours) matched the initial estimates. Much of the time was allocated to the new Change Distribution Module (1 hour) to ensure it complies with the specific requirements and constraints.

4. Realized risks and newly discovered risks.

- *Mitigated Risks:* The critical risks identified in the previous report (lack of API request chunking and hardcoded date limitations) have been successfully resolved. The system now correctly chunks requests and provides user-friendly error boundaries.
- *Newly Discovered Risks:* A critical business logic risk has been identified in the new module. The application calculates and displays changes for two currencies independently instead of calculating the cross-rate for the selected currency pair as mandated by the specification. This makes the current analytical output mathematically incorrect from a business perspective.

5. Conclusions for the future.

- The re-testing phase confirmed that the development team successfully addressed previous issues, highlighting the effectiveness of detailed bug reporting.
- The critical logic error found in the Change Distribution Module emphasizes the need for developers to closely consult the Project Specification regarding business logic (e.g., cross-rate calculations) before implementation.